

FJM Sequence of Operation

Operating Range:

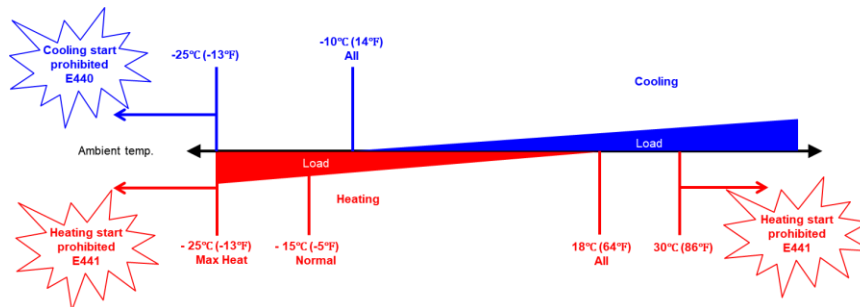
Cooling above 50°F if out of range an E440 error will be displayed, Cooling Start Prohibit.

Heating:

Heat Pump 5°F - 64°F (86°F)

Max Heat -13°F - 64°F (86°F)

If out of range a E441 will be displayed, Heating Start Prohibit.



Thermo On - Off:

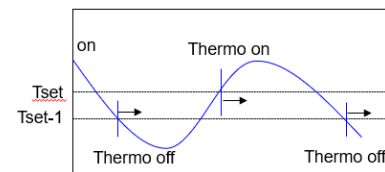
Cooling:

JFM- On when the space is above setpoint.

Off when temp drops 1.8 degrees below setpoint.

RAC - On when the space is 1.8°F above setpoint.

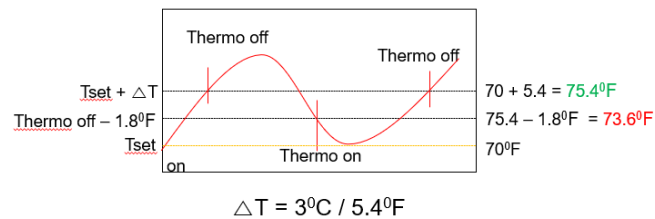
Off when temp drops below setpoint.



Heating: On : Room temp. $\leq$ Set temp. $- 1.8^\circ\text{F} + \Delta T$

Off : Room temp. $\geq$ Set temp. $+ \Delta T$

($\Delta T^\circ\text{C}$: Heating compensation temperature)



* Segment 6 of the product option code can be used to adjust the

Compressor Capacity Control:

Compressor Hz is determined using outdoor temperature and ΔT .

(ΔT is difference of between the room temperature and set temperature.)

When the room temperature approaches the set temperature, the compressor Hz reduced according to “Fuzzy control”

Cooling:

Fuzzy control

Use Room temp. & target temp of running units

Gap is big : Hz up

Gap is small : Hz down

“Typical evaporator temperature” is between 40°F and 45°F.

Low side pressure 118 psig – 130 psig.

Heating:

Target IDU temperature control

* Target: 47~52°C (117~126°F) (only 1 IDU on = target -5°C(9°F))

High side pressure 402 psig - 453 psig

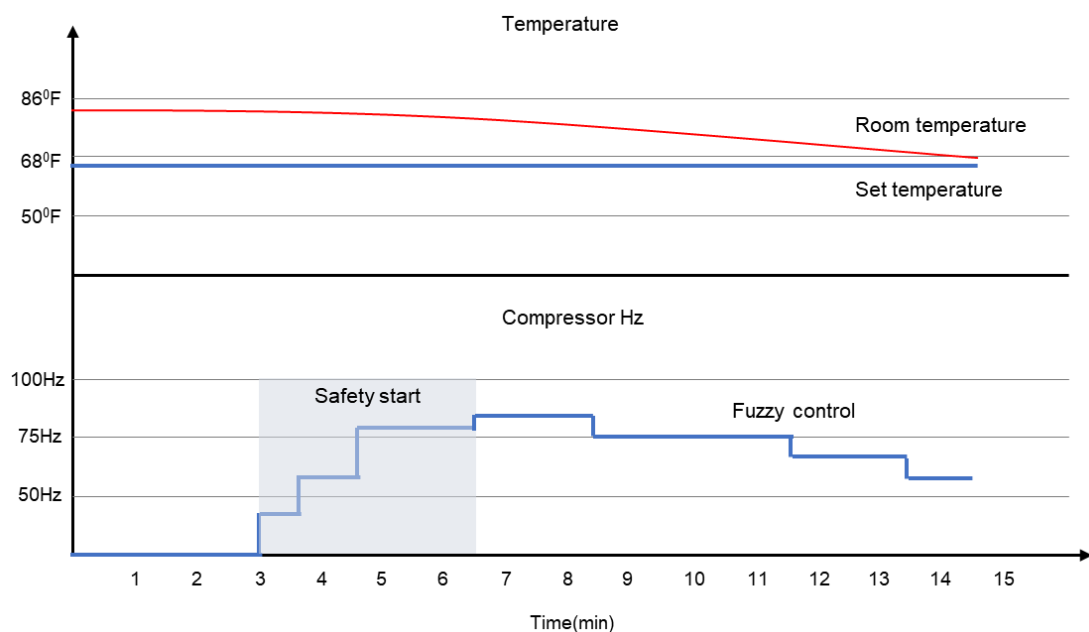
* At this time we use calculated IDU temperature as a target

Calculate average IDU temperature of running indoor units

Average temperature < target : Hz up

Average temperature > target : Hz down

Average temperature - target $\leq 1^{\circ}\text{C}(2^{\circ}\text{F})$: no change



Freeze Protection (FJM):

To prevent freezing of the indoor coils in cooling mode.

Freeze protection **initiation**:

The lowest temperature amongst, evap in / evap mid / evap out / pipe in / pipe out is used to initiate freeze protection.

Freeze protection **termination**:

The evap. in / mid / evap. out thermistors are used to terminate freeze protection.

1st Freeze protection:

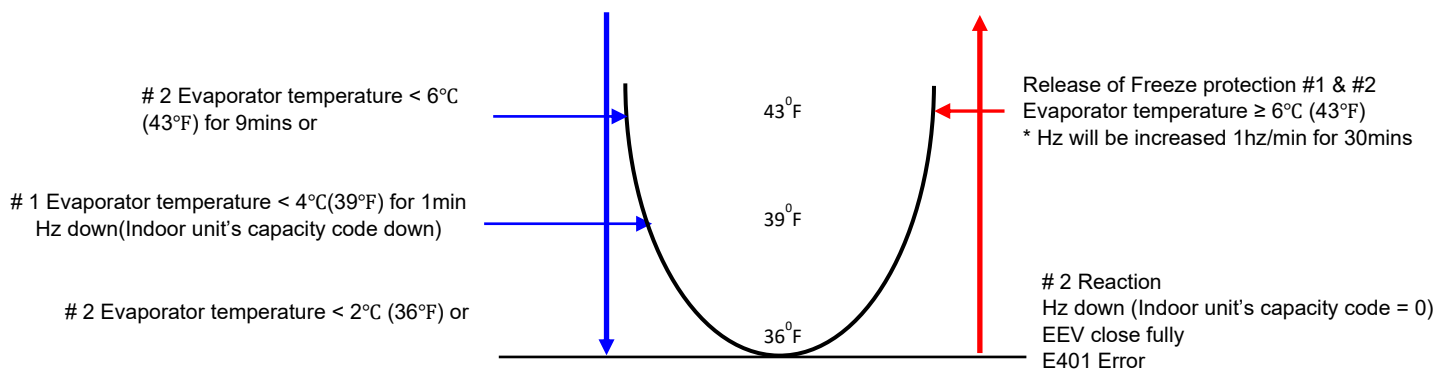
If Evaporator temperature is $< 4^{\circ}\text{C}$ (39°F) for 1 minute then the compressor Hz and the Indoor unit's capacity code down will be reduced

2nd Frost protection:

If the evaporator temperature $< 2^{\circ}\text{C}$ (36°F) **OR** is $< 6^{\circ}\text{C}$ (43°F) for 9 minutes then the compressor Hz will be reduced, the Indoor unit's capacity code will go to 0 and the EEV will fully close and an E401 error will be displayed.

Release of conditions for 1st or 2nd protections:

When the Evaporator temperature is $\geq 6^{\circ}\text{C}$ (43°F) (Use evap in/mid/out for release), Freeze protection will be released and the compressor Hz will be increased 1hz/min for 30mins.



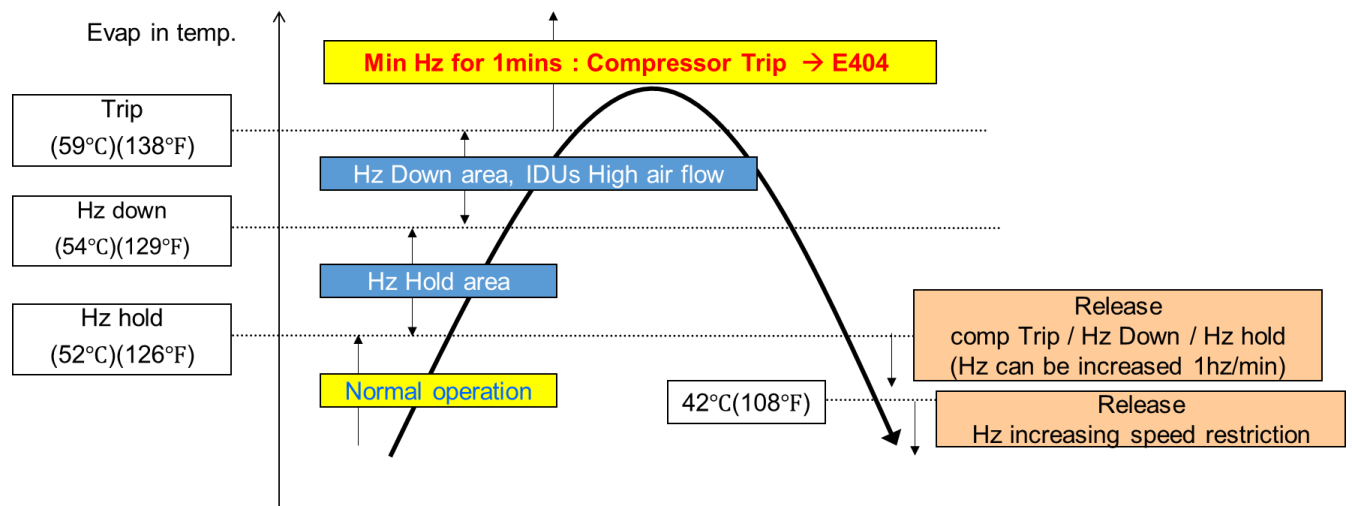
Heating Overload Protection:

To prevent overheating of indoor coil in the heating mode.

The evap. in thermistor is used for judgment.

Because there is a direct relationship between the system temperature and pressure. These temperatures can be used to get an approximate high side operating pressure range.

The system high side pressure measured at the gas line connection should be within the range of 355psig (108°F) and 470psig (129°F).



Discharge Temperature Protection:

To protect the compressor from overheating.

Target discharge temperature is determined using the compressor Hz, Room temp and ambient temp.

The target discharge temperature can be read using the K 4 Button. (press K 4, 14 times).

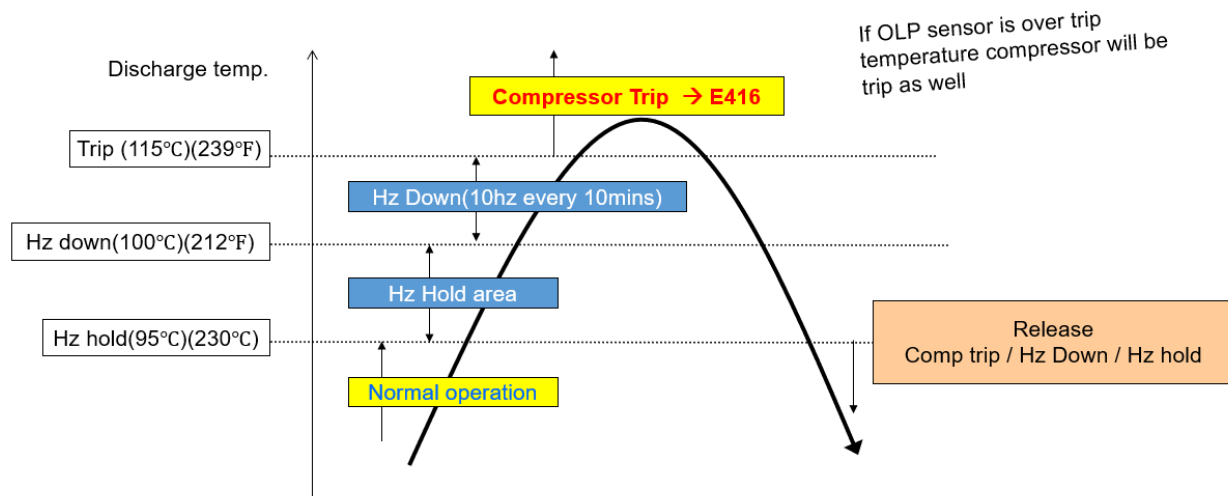
The discharge temperature is directly related to the evaporator superheat. If the evaporator superheat is low, then the compressor discharge temperature will be low. If the evaporator superheat is high, then the discharge temperature will be high.

The IDU EEV's are used to regulate the discharge temperature.

The Discharge temperature and OLP Temperature can be read using the K 4 button.

To read the Discharge Temperature - Press the K 4 button 9 times.

To read the OLP Temperature - Press the K 4 button 10 times.



EEV / Capacity Control:

The EEV is adjusted to match the indoor units' output to the operating load.

It's primary purpose is discharge temperature control. This is based on the direct relationship between evaporator superheat and discharge superheat.

Evaporator superheat is a factor of the system charge and load.

On start up the EEV will be driven to an initial setting based on the size of the indoor unit.

After start up its operation is as follows:

Cooling Mode:

1. Target discharge control
2. Evap. super heat control(target 1~4°C(2~7°F))

SH high = EEV open / SH low = EEV close

When Indoor unit is thermally satisfied or powered off the EEV is driven closed (0 steps)

Heating Mode:

1. Target discharge control
2. Evaporator in temperature control
 - Current temperature < average : EEV open
 - Current temperature > average : EEV close

Provides refrigerant balance between each units

When Indoor unit is thermally satisfied the EEV is driven to 80 steps to prevent refrigerant. Stagnation. When powered off the EEV is driven closed (0 steps)

When a new indoor unit is turned on the IDU's (all IDUs) EEV step will go back to its initial step setting.

This is done to secure system capacity & reliability. If the EEV step is maintained, when new IDU is turned on, there will be more refrigerant flow to certain IDU 's required more time for the system to balance out.

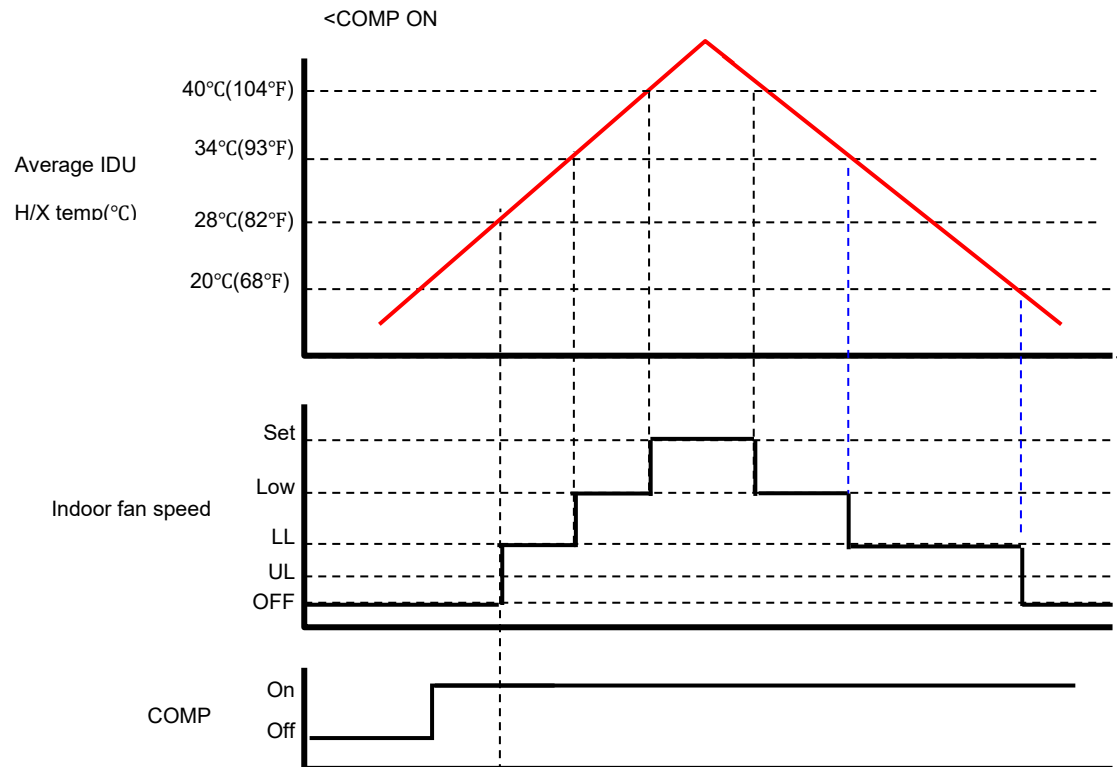
When power is applied to the unit the EEV will be driven 50 steps past fully closed (530 steps). There will be a short delay and then the EEV will be driven fully open (480 steps).

Cold Air Prevention:

Control the fan speed to avoid discomfort of people in the room due to cold air during the initial start-up of heating

The fan does not operate if the indoor heat exchanger temperature is below 68°F in heating mode.

When the Indoor units are operating in heating mode the fan speed is restricted according to the average heat exchanger temperature and temperature variations (increase/decrease).



Control value may vary by each model (Average IDU H/X temp. = average of (evap in /evap out))

Compressor Preheat / Crankcase heater:

To protect the compressor from failure due to oil forming

Ratio of Oil diluting is increased at low temperature → High possibility of oil forming.

Operating condition (A x B x C x D)

A : Compressor = 0Hz

B : Ambient temperature $\leq 0^{\circ}\text{C}$

C : OLP temperature $\leq$ Ambient temperature + 10°C (18°F)

D : 10min after Compressor stop

Release Condition (A or B or C or D)

A: Compressor $\neq$ 0Hz

B: Ambient temperature $\geq 2^{\circ}\text{C}$ (36°F)

C: OLP temperature $\geq 10^{\circ}\text{C}$ (50°F)

D: Motor coil heating time $\geq 180\text{mins}$

High Pressure Blockage Control (E422)

During operation check sensing conditions to sense the high pressure.

Set off and display the error(E422)

If all Sensing conditions are met (1, 2 & 3) after 4 minutes of compressor operation the unit is cycled off and a E422 error is displayed

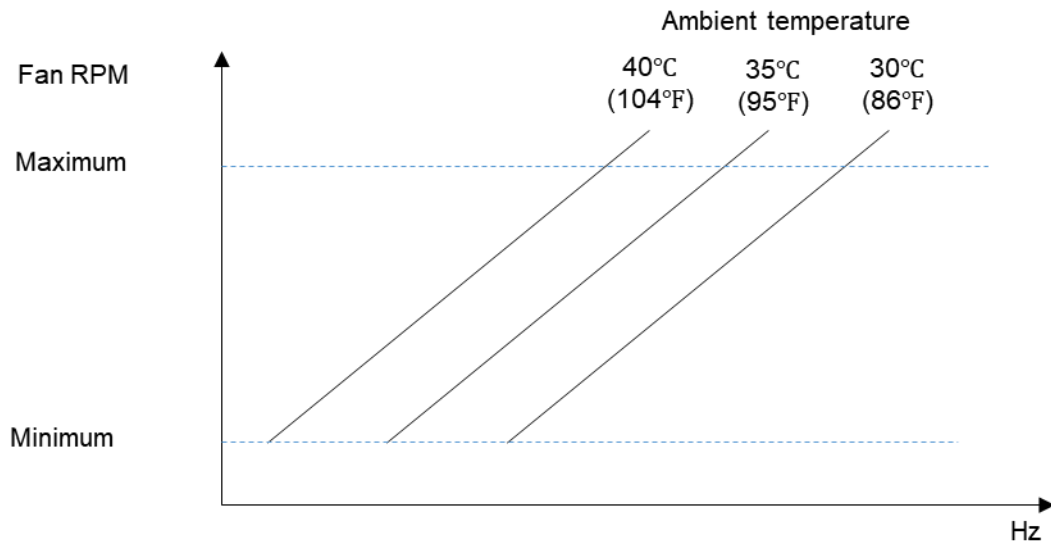
Sensing condition	
1	Room temp. – Evap in temp. $\leq 3^{\circ}\text{C}$ (5°F)
2	Room temp. – Evap mid temp $\leq 3^{\circ}\text{C}$ (5°F)
3	Current Comp Hz $\geq$ Hold 1 Hz

Outdoor Fan Speed Control:

Function of (current compressor frequency & ambient temperature)

High Hz → high speed

High ambient temperature → High speed (cooling) / Low speed (Heating)



Dew prevention control (For Wind-Free model):

To prevent condensation from forming on the product.

Judgment: Starts after 35mins of Wind-Free operation

A condition : $60\% \leq RH \leq 70\%$, RH : Relative Humidity

Then, run the dew removal operation according to the installation option SEG 12.

B condition : $RH \geq 70\%$

Then, run the dew removal operation according to the installation option SEG 12.

Release Condition (A or B or C or D)

A: Not Wind-Free mode

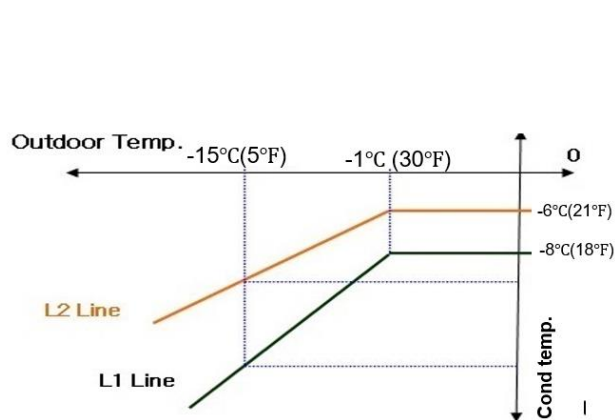
B: When thermos off condition

C: Not Cooling mode

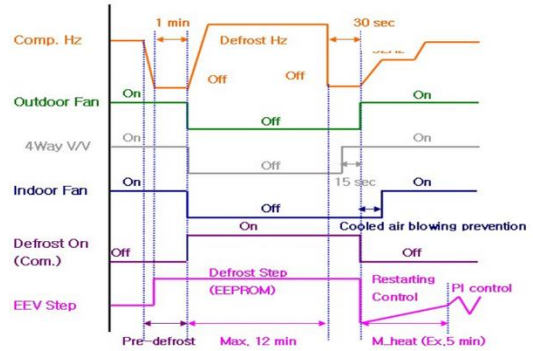
D: $RH < 55\%$

Defrost Operation:

To remove the ice on the outdoor unit's heat exchanger to secure heating capacity



EEV full open during defrost



- Defrost operation condition

: (①x②) or (①x③x④)

- ① Heating mode (Comp. operating)
- ② OD HX temp. under L2 line 120min (accumulated)
- ③ Comp. On during 35 min
- ④ OD HX temp. under L1 line during 3 min

- Release condition(① or ②)

- ① Comp. On during 12 min in defrost operation
- ② OD HX temp. over Temp. released (Ex. 15°C(59°F))

Frequencies

Model #	Cf_C_Autoa_Nom Hz	Min Heat	Max Heat	Min Cool	Max Cool
AJ020BXJ2CH	45	15	100	15	47
AJ024BXJ3CH	45	15	85	15	50
AJ036BXJ4CH	55	15	100	15	47
AJ048BXJ5CH	55	15	115	15	85
AJ020BXS3CH	55	15	110	15	65
AJ024BXS4CH	55	15	110	15	65
AJ030BXS4CH	55	15	110	15	65
AJ036BXS4CH	55	15	115	15	65

E554

E554 Inspection:

- *Judgement begins once compressor frequency is higher than EEPROM programmed value.*
 - *If compressor frequency falls below EEPROM programmed value, detection period resets (starts over)*
 - *If compressor turns off during the detection period, detection timer will reset (starts over)*
- *Compressor operation must maintain the frequency value higher than the EEPROM programmed frequency value for at least 7 minutes continuously.*
- *Refrigerant leakage judgment conditions:*
 - *Only indoor units that are powered on and in a thermal on state for 7 minutes are part of the E554 inspection.*
 - *Refrigerant leakage is determined if the below conditions are met for 5 minutes continuously, starting at the beginning of the inspection initiation:*
 - *During cooling operation:*
 - *Return Air temp – Eva in temp < 4°C (7.2°F)*
 - *Return Air temp – IDU mid sensor temp < 4°C (7.2°F)*
 - *Cond out temp – outdoor air temp < 3°C (5.4°F)*
 - *During heating operation:*
 - *Eva in temp – return air < 2°C (3.6°F)*
 - *IDU mid sensor – Return air < 2°C (3.6°F)*
 - *Cond out temp – outdoor temp ≥ -2°C (-3.6°F)*
- *Non-leak decision: If any IDU does not meet the above conditions for 5 consecutive inspections, the error will not populate.*
- *Repeat inspection: Refrigerant detection will not be performed again until power supply is reset.*
- *Clearing of results: If the pipe check operation is restarted, the previous refrigerant leakage detection judgement will be cleared, and the system will be reinspected when criteria is met.*
- *In the case of normal judgment, the operation status will be maintained.*
- *Refrigerant leak detected: E554 displayed.*